Solution: Exercise 1 - Work and Energy

System Parameters & Given Data

A point mass (S), of mass $m = 250$ g, is released without initial velocity from point A of an ABCD track located in the vertical plane.

- Friction is negligible on parts AB and CD of the track.
- The friction on part BC is equivalent to a constant force of magnitude $f = 1$ N.
- Given angles and dimensions: $\alpha = 45°$, $AB = BC = 1$ m, and $g = 10$ m/s^2.
- We choose the horizontal plane passing through B and C as the reference level for the Gravitational Potential Energy ($GPE = 0$).

Preliminary Conversions

Mass $m = 250$ g $= 0.25$ kg.
Initial velocity at A: $v_A = 0$ m/s.

Question 1.a: Speed at B

Apply the Kinetic Energy Theorem to mass (S) between A and B:

$$\Delta KE_{A\to B} = \sum W(\vec{F})$$

$$KE_B - KE_A = W_{A\to B}(\vec{W}) + W_{A\to B}(\vec{N})$$

Since the mass is released from rest, $KE_A = 0$. The normal force $\vec{N}$ is perpendicular to the displacement at all times, so $W(\vec{N}) = 0$.

$$\frac{1}{2}mv_B^2 = mgh_{AB}$$

The vertical drop h_{AB} is determined by the length AB and angle α:

$$h_{AB} = AB \cdot \sin(45°) = 1 \cdot \frac{\sqrt{2}}{2} \approx 0.707 \text{ m}$$

Solving for v_B:

$$v_B = \sqrt{2gh_{AB}} = \sqrt{2 \cdot 10 \cdot 0.707} = \sqrt{14.14} \approx 3.76 \text{ m/s}$$

Question 1.b: Speed at C

Apply the Kinetic Energy Theorem to mass (S) between B and C. The acting forces are weight $\vec{W}$, normal force $\vec{N}$, and friction $\vec{f}$:

$$\Delta KE_{B\to C} = W_{B\to C}(\vec{W}) + W_{B\to C}(\vec{N}) + W_{B\to C}(\vec{f})$$

Since the track BC is horizontal, the work done by weight and normal force is zero. The friction opposes motion:

$$\frac{1}{2}mv_C^2 - \frac{1}{2}mv_B^2 = -f \cdot BC$$

$$v_C = \sqrt{v_B^2 - \frac{2f \cdot BC}{m}}$$

Substitute the known values ($v_B^2 = 14.14$ m^2/s^2):

$$v_C = \sqrt{14.14 - \frac{2 \cdot 1 \cdot 1}{0.25}} = \sqrt{14.14 - 8} = \sqrt{6.14} \approx 2.48 \text{ m/s}$$